Assignment 1: Report

November 3, 2026

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1.1 MLP

Started with a baseline model with hidden_size = 256, dropout = 0.2, optimizer = SGD, and learning_rate = 0.01. Used a batch of 64 and 10 epochs. Here, we varied the activation layers. Tried out ReLU, LeakyReLU, and GELU. Using a 90/10 split for training and validation, LeakyReLU resulted in the highest validation accuracy = 0.9018.

Will continue varying the optimizers, using LeakyReLU as the activation function. Tried out SGD, Adam, and AdamW. Again using a 90/10 split for training and validation, AdamW resulted in the highest validation accuracy = 0.8650.

Learning rate
Depth/width
Dropout

1.2 CNN